Internals Know More Than the Model Can Say: Relevance Signals Are Encoded Internally but Attenuated en Route to the Output

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Abstract

We study whether large language models (LLMs) "know" more about retrieval relevance than their outputs reveal. Using linear probes on a single frozen forward pass over (query, passage) pairs, we compare four relevance *readers* on the same model, prompt, and answer token: (i) the model's literal output verdict (zero-shot LLM-judge), (ii) a trained probe on the final-layer hidden state (the *output representation*), (iii) a trained probe on the best residual-stream layer (*internal*), and (iv) a probe on the top attention heads. Across six models (3B–8B, LLaMA and Mistral, base and instruct), reading internal activations decodes relevance markedly better than reading the output. With query-clustered bootstrap confidence intervals, the attention-head probe beats the output-representation probe on all four LLaMA models (each $p < 10^{-4}$); after scaling the evaluation to tighten power, the residual-stream advantage over the output representation is individually significant on base models (+0.077, p=0.0008; +0.060, p=0.004). A causal steering intervention confirms the link is mechanistic, not epiphenomenal: ablating the internal relevance direction at the output layer monotonically collapses the model's verdict (mean P(yes) 0.79→0.26), while a random direction and the output's own direction fail to preserve discriminative AUC. Comparing base and instruct pairs, we find alignment improves the output readout (the judge gets stronger) yet does not close the internal–output gap. In short: the relevance signal is encoded internally and *attenuated*, not absent, on the path to the logits.

1 Introduction

When an LLM is asked whether a retrieved passage is relevant to a query, its literal answer is often little better than chance. Yet the same forward pass may already contain a far cleaner relevance signal in its internal activations. We ask a focused, mechanistic question: *is relevance better decodable from a model's internals than from its output, and if so, is the internal signal causally connected to the output the model fails to produce?*

This is an interpretability question, not a retrieval-engineering one. We do **not** claim a probe that beats a strong cross-encoder reranker, nor zero-shot cross-domain robustness; prior exploration refuted those framings. Our contribution is the *relative* science of internals vs. output:

- **Existence.** On the same model/prompt/answer-token, a linear probe on attention-head activations decodes (query, passage) relevance substantially better than the model's own output verdict, and better than a trained probe on the output representation (final hidden state).

- **Attribution.** Decomposing the gap into a *training* component (final-layer probe − judge) and a *depth/internal* component (best internal layer − final layer) shows the edge comes substantially from *reading internals*, not merely from *being trained*.
- **Mechanism.** Relevance decodability peaks in the middle of the network and declines toward the output layer. Comparing base vs. instruct models, alignment *improves* the output readout but does not eliminate the internal–output gap.
- **Causality.** A steering intervention that injects the internal relevance direction at the output layer causally controls the verdict; ablation is cleanly monotonic and, unlike control directions, preserves discriminative AUC—evidence that the output pathway *can* express the signal but normally under-expresses it.

We report query-clustered bootstrap confidence intervals throughout, and are explicit about which claims are individually significant, which rest on a cross-model sign test, and which are architecture-dependent (the attention-head reader is weak on Mistral). Honest delineation of effect strength is central to this paper.

2 Related Work

Probing classifiers read latent features from frozen representations to test what information is linearly available [Alain and Bengio, 2017, Hewitt and Manning, 2019]; we use them comparatively across depth and against the output. **Logit lens and tuned lens** [nostalgebraist, 2020, Belrose et al., 2023] project intermediate states to the vocabulary, motivating our depth analysis. **Activation steering / representation engineering** [Turner et al., 2023, Zou et al., 2023, Li et al., 2023] edit residual directions to change behavior; we use a difference-of-means steering vector for a causal test. **Calibration and hallucination** work shows models can encode correctness they do not surface [Kadavath et al., 2022, Azaria and Mitchell, 2023]; our finding is the relevance-specific, depth-resolved analogue. **RAG relevance** estimation usually targets reranking accuracy; we deliberately stay on the interpretability question and do not compete with rerankers.

3 Method

Task and data. We use MS MARCO v1.1, forming (query, passage, label) triples with relevance labels. Splits are made *by query* (70/15/15) so no query appears in both train and test, preventing leakage. We report ROC-AUC on the held-out test set.

Prompt and extraction. Every reader sees the identical instruct prompt, `"Document: {p}\nQuestion: {q}\nIs the document relevant to the question? Answer:"`, and all activations are taken at the final *answer token*. Per-head activations are the `o_proj` outputs reshaped to $[\text{heads}, d_{\text{head}}]$; the residual stream is each decoder layer's output.

Four readers. (1) *LLM-judge*: the model's own normalized $P(\text{yes})$ vs. $P(\text{no})$ at the answer token, *zero training*. (2) *Output-representation probe*: an ℓ_2-regularized logistic regression on the final hidden state. (3) *Internal probe*: the same probe on the best residual-stream layer, selected *on validation*. (4) *Attention-head probe*: top-k heads chosen on validation, their activations concatenated into one logistic regression.

Gap decomposition. We split the total edge over the judge into train_edge = final − judge (pure training, both at the output) and internal_edge = best_resid − final (pure depth, beyond

Table 1: Four relevance readers across six models (test AUC, point estimates). *judge*: output verdict, no training; *final*: probe on output representation; *best-resid*: probe on best internal layer; *attn*: top-head probe. Significance in Tables 2–3.

Model	L	judge	final	best-resid	attn
LLaMA-3.2-3B (base)	28	0.535	0.593	0.632	**0.685**
LLaMA-3.1-8B (base)	32	0.514	0.552	0.634	**0.680**
Mistral-7B-v0.3 (base)	32	0.531	0.602	**0.671**	0.603
LLaMA-3.2-3B-Instruct	28	0.597	0.607	0.704	**0.725**
LLaMA-3.1-8B-Instruct	32	0.603	0.585	0.666	**0.746**
Mistral-7B-Instruct-v0.3	32	0.621	0.618	0.662	**0.667**

Table 2: Query-clustered bootstrap (5000 resamples), 500-query evaluation. Edges with 95% CI and one-sided p. **Bold**: $p < 0.05$.

Model	internal_edge	attn−final	best_resid−judge
LLaMA-3.2-3B	+0.036 [-.05,.12]	+0.094 [.01,.18]	+0.093 [.01,.18]
LLaMA-3.1-8B	+0.028 [-.06,.11]	+0.128 [.04,.22]	+0.065 [-.02,.15]
Mistral-7B	+0.019 [-.07,.10]	+0.002 [-.08,.09]	+0.089 [.00,.17]
LLaMA-3.2-3B-Inst	+0.026 [-.06,.11]	+0.119 [.03,.21]	+0.035 [-.05,.13]
LLaMA-3.1-8B-Inst	+0.044 [-.04,.13]	+0.164 [.08,.25]	+0.024 [-.06,.11]
Mistral-7B-Inst	+0.011 [-.08,.10]	+0.051 [-.04,.15]	+0.006 [-.08,.09]
sign test (p)	6/6 (**.016**)	6/6 (**.016**)	6/6 (**.016**)

training). The thesis "internals > output" requires internal readers to exceed *both* the judge and the output-representation probe.

Significance. Because the split is by query, the correct resampling unit is the query. We report 95% confidence intervals and one-sided p-values from a *query-clustered bootstrap* (5000 resamples) on frozen test predictions; the best layer is chosen on validation to avoid selection-on-test optimism. We additionally run a cross-model sign test on the direction of each edge.

Causal steering. At the best internal layer we form a probe-free relevance direction $d = \overline{h}_{\text{rel}} - \overline{h}_{\text{irrel}}$ (difference of class means, unit-normalized). During the forward pass we add $\alpha\|h\|d$ to the *output-layer* residual at the answer token and sweep α, reading the judge's $P(\text{yes})$ and the test AUC. We compare against the output layer's own difference-of-means direction and a random unit direction (norm-matched).

4 Experiments

Internals decode relevance better than the output. Table 1 reports the four readers across six models. On every model the best internal reader exceeds both the zero-shot judge and the trained output-representation probe. The judge itself hovers near chance on base models (AUC 0.51–0.54), confirming the output "says" little.

Significance. Table 2 gives query-clustered bootstrap CIs. The *attention-head vs. output-representation* edge is individually significant on all four LLaMA models ($p < 10^{-4}$, effect +0.09 to +0.16). The residual-stream edge (internal_edge) is directionally positive on all six models (sign test p=0.016) but, at ∼75 test queries, not individually significant—a power problem, not an absent effect.

Table 3: Scaled (1500-query) re-test of internal_edge vs. the 500-query estimate. **Bold**: $p < 0.05$.

Model	internal_edge (500q)	internal_edge (1500q)
LLaMA-3.2-3B (base)	+0.036, p=0.19	+0.077 **[.03,.12],** p=0.0008
LLaMA-3.1-8B (base)	+0.028, p=0.26	+0.060 **[.02,.10],** p=0.004
LLaMA-3.2-3B-Inst	+0.026, p=0.28	+0.024 [-.02,.07], p=0.15
LLaMA-3.1-8B-Inst	+0.044, p=0.16	+0.012 [-.04,.06], p=0.32

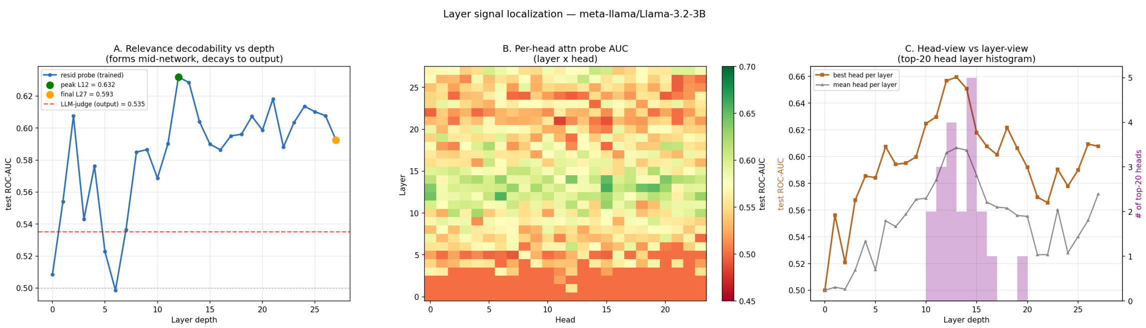

Figure 1: Depth localization (LLaMA-3.2-3B). (a) per-layer residual probe AUC vs. depth with the judge baseline and the peak/output markers; (b) per-head AUC heatmap; (c) head- vs. layer-view and the layer distribution of the top heads. Signal forms mid-network and attenuates toward the output.

Scaling the evaluation confirms the internal edge. We re-extract four flagship models at 1500 queries (∼225 test queries; sample size is the only change) and re-test (Table 3). The residual-stream edge becomes individually significant on base models (+0.077, p=0.0008; +0.060, p=0.004): the p-values drop from the 0.2 range to the 0.001 range purely from added power. Strikingly, the edge is significant on base models but shrinks and loses significance on instruct models—a mechanistic signal we return to below.

Depth localization. Figure 1 (per model in the supplement) shows relevance decodability rising into the mid network and declining toward the output layer; the judge baseline sits well below the peak. For LLaMA the strongest heads cluster in the middle (L10–16); for Mistral single heads carry little signal (early-layer head AUC $\approx$ 0.50) while the residual stream still encodes relevance—a real architectural difference, not a failure of the thesis.

Mechanism: base vs. instruct. Across three base→instruct pairs, alignment shows a consistent triple effect: the judge improves (output "says" more), train_edge falls toward or below zero (training a final-layer probe adds little once aligned), and the internal edge persists (and widens on LLaMA). Combined with Table 3 (the internal edge is significant on base, compressed on instruct), the reading is: *alignment improves the output readout but does not close the internal–output gap*—it does not merely "suppress" the output.

Causality. Figure 2 shows the steering sweep. Injecting the internal relevance direction at the output layer drives the verdict monotonically on the ablation arm (P(yes) 0.79 → 0.26 as α<0), and is the *only* direction that preserves (even slightly improves) discriminative AUC (0.546 → 0.553). The output's own direction moves P(yes) strongly (0.01 → 0.99) but *destroys* AUC (→ 0.485, a uniform logit shift), and a random direction leaves AUC flat. The additive arm (α>0) saturates because the judge is already yes-biased (baseline 0.79 while only ∼22% of test pairs are relevant), so we base the causal claim on the ablation arm and the AUC contrast. This indicates the output

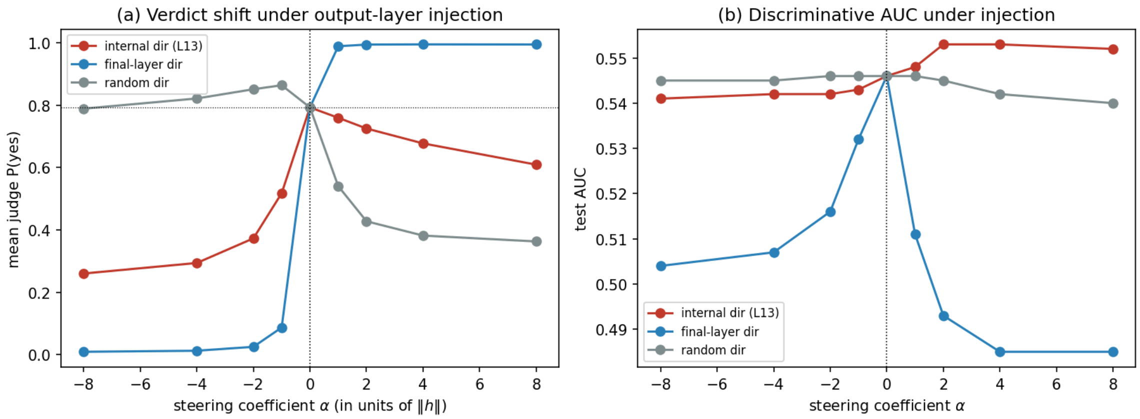

Figure 2: Causal steering at the output layer (LLaMA-3.2-3B). (a) mean judge $P(\text{yes})$ vs. steering coefficient α; (b) test AUC. Only the internal direction preserves AUC; the final-layer direction inflates $P(\text{yes})$ while collapsing AUC.

pathway *can* express the internal relevance signal but normally under-expresses it: attenuation, not absence.

5 Discussion and Limitations

The relevance signal an LLM needs to judge a passage is present in its internals and only partially surfaced at the output; alignment tightens the readout without closing the gap. This connects to calibration/hallucination findings that models "know" more than they assert, localized here to a specific signal and resolved over depth, with a causal handle.

Limitations. (1) Probing is correlational; our causal evidence is a single-model steering study (LLaMA-3.2-3B) and its additive arm is limited by the judge's yes-bias. (2) We use one in-domain dataset (MS MARCO); we make no cross-domain or reranking-accuracy claims, which prior exploration refuted. (3) The attention-head reader is architecture-dependent (weak on Mistral), so we treat the residual-stream layer as the canonical "internal" reader and the head probe as a stronger LLaMA-family instrument. (4) Absolute AUC values are modest (0.5–0.75); the contribution is the *relative* internal-vs-output relationship, not a state-of-the-art relevance classifier.

6 Conclusion

On the same model, prompt, and token, relevance is more decodable from internal activations than from the output the model produces, the advantage comes substantially from reading depth rather than from training, the signal peaks mid-network and attenuates toward the logits, and a steering ablation shows the internal direction causally underlies the verdict. Internals know more than the model can say.

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